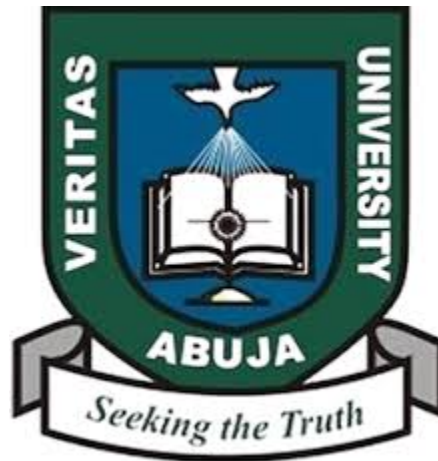


Department of Software Engineering
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‘LAB MANUAL’
Course Code: VUA-SEN182
Course Title: “Web Programming (HTML, CSS, Git)”

Prepared by: Mr. Victor Ayamba

Course Contents

Contents	CLOs	Lecture	Practical
HTML: Basics, Elements, Attributes, Headings, Paragraphs, Formatting, Links, Head, Images, Tables, Lists, Blocks, Layout, Forms, Multimedia Tags, Semantic HTML, HTML5 Structure.	CLO – 1 & 2	4	8
CSS: Introduction, Syntax, Selectors (ID & Class), Styling Backgrounds, Text, Fonts, Links, Box Model (Border, Outline, Margin, Padding), Grouping/Nesting, Dimension, Display, Positioning, Floating, Align, Navigation Bar, Image Gallery, Flexbox, Grid Layout, Responsive Design.	CLO – 1 & 2	3	6
JavaScript: Introduction, Output, Statements, Comments, Variables, Data Types, Objects, Functions, Operators, Comparisons, Conditions, Loops, Error Handling, DOM Introduction and Manipulation, Number, String, Date, Array, Window, Screen, Navigator, Popup Alerts.	CLO – 1 & 2	4	8
Bootstrap: Introduction to Bootstrap Framework, Grid System, Containers and Layout, Responsive Design Principles, Typography and Utilities, Navigation Bars, Buttons, Cards, Forms and Input Groups, Tables, Alerts, Modals, Responsive Components and Layout Customization.	CLO – 1 & 2	4	8
Total		15	30

INTRODUCTION TO THE COURSE:

The Internet and the World Wide Web have become fundamental components of modern society, influencing communication, education, business, and everyday life. As digital platforms continue to expand, the ability to design and develop functional and visually appealing web applications has become an essential skill for software engineers and technology professionals. Understanding how web technologies work together enables students to create interactive systems that can serve real-world needs.

This laboratory course provides a practical introduction to modern Web Programming using core client-side technologies such as **HTML, CSS, JavaScript, and the Bootstrap framework as well as version control tools like Git and Github**. Students will learn how to structure web pages using HTML, style and layout interfaces with CSS, enhance interactivity through JavaScript, and build responsive, mobile-friendly user interfaces using Bootstrap. Through hands-on laboratory exercises, students will progressively develop the skills required to design, implement, and maintain modern web applications.

The lab sessions are structured to guide students step-by-step through the development of a meaningful web project while introducing essential web development concepts and tools. Each experiment builds upon the previous one, enabling learners to gradually transform basic web pages into fully structured and responsive websites. At the end of each lab session, tasks are provided to reinforce understanding and encourage independent exploration. Students are advised to follow this lab manual sequentially and carefully adhere to the provided guidelines to gain maximum benefit from the course.

Lab 3: HTML Colors, CSS, Links, Images, FavIcon, Page Title, Lists, HTML Block and Inline - Demonstrated by Instructor

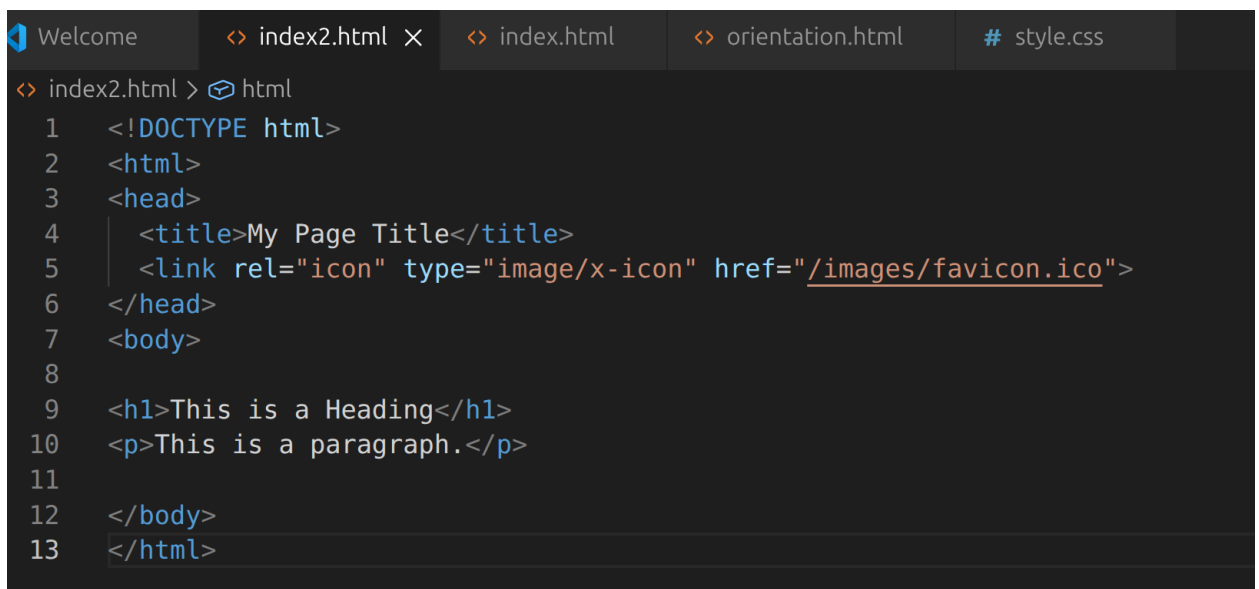
Experiment Name: HTML Fundamentals; Document Structure, Text Elements, Links, Images and Basic Styling

FavIcons in HTML

A favicon is a small image displayed next to the page title in the browser tab. You can use any image you like as your favicon. You can also create your own favicon on sites like <https://www.favicon.cc>.

To add a favicon to your website,

1. Save your favicon image to the root directory of your website or
2. Add a folder in the root directory called images, and save your favicon image in this folder.
3. A common name for a favicon image is "favicon.ico".
4. Next, add a **<link>** element to your "index.html" file, after the **<title>** element, like this as shown on line 5



```
Welcome  index2.html x  index.html  orientation.html  # style.css
index2.html > html
1  <!DOCTYPE html>
2  <html>
3  <head>
4      <title>My Page Title</title>
5      <link rel="icon" type="image/x-icon" href="/images/favicon.ico">
6  </head>
7  <body>
8
9      <h1>This is a Heading</h1>
10     <p>This is a paragraph.</p>
11
12     </body>
13 </html>
```

5. Now, save the "index.html" file and reload it in your browser. Your browser tab should now display your favicon image to the left of the page title.

FavIcon Formats

The following table shows the file format support for a favicon image:

Browser	ICO	PNG	GIF	JPEG	SVG
Edge	Yes	Yes	Yes	Yes	Yes
Chrome	Yes	Yes	Yes	Yes	Yes
Firefox	Yes	Yes	Yes	Yes	Yes
Opera	Yes	Yes	Yes	Yes	Yes
Safari	Yes	Yes	Yes	Yes	Yes

Figure 2: FavIcon Formats. Source [W3School](#)

1. Open a code editor such as **Visual Studio Code**, Notepad++, or Sublime Text.
2. Create a new file.
3. Save the file with the extension **.html** (example: **index.html**).
4. Open the file using a web browser.

HTML Page Title

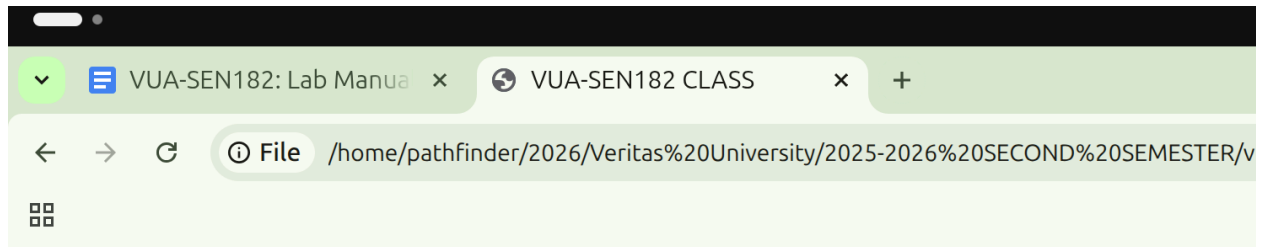
Every web page should have a page title to describe the meaning of the page. The `<title>` element adds a title to your page:

Example

```
<!DOCTYPE html>
<html>
<head>
  <title>VUA-SEN182 CLASS</title>
  <link rel="icon" type="image/x-icon" href="/images/favicon.ico">
</head>
<body>

</body>
</html>
```

Example



Welcome to the VUA-SEN182 CLASS!

Figure 3: Web Page with a Title Page

Note: The title should describe the content and the meaning of the page. The page title is very important for search engine optimization (SEO). The text is used by search engine algorithms to decide the order when listing pages in search results.

HTML Tables

HTML tables allow web developers to arrange data into rows and columns.

Example:

Welcome to the VUA-SEN182 CLASS!		
Student Name	Student ID	Email
John Doe	123456	john.doe@example.com
John Doe	123456	john.doe@example.com

Figure 4: HTML Table

Table Cells

- Each table row starts with a `<tr>` and ends with a `</tr>` tag.
- `tr` stands for table row
- You can have as many rows as you like in a table; just make sure that the number of cells are the same in each row.

Example:

```
<tr>
  <td>Emil</td>
  <td>Tobias</td>
  <td>Linus</td>
</tr>
<tr>
  <td>16</td>
  <td>14</td>
  <td>10</td>
</tr>
```

Table Cells

- Each table cell is defined by a **<td>** and a **</td>** tag.
- **td** stands for table data.
- Everything between **<td>** and **</td>** is the content of a table cell.

Example Structure:

```
</tr>
  <tr>
    <td>John Doe</td>
    <td>123456</td>
    <td>john.doe@example.com</td>
  </tr>
```

HTML Paragraphs

Paragraphs are defined using the **<p>** tag.

Example:

```
<p>This is a paragraph.</p>
<p>This is another paragraph.</p>
```

Important characteristics:

- A paragraph always starts on a **new line**.
- Browsers automatically add **spacing before and after paragraphs**.

Table Headers

- If there is a need to make cells to be table header cells. In those cases use the `<th>` tag instead of the `<td>` tag:
- By default, the text in `<th>` elements are bold and centered, but you can change that with CSS.

Example:

```
<table>
  <tr>
    <th>Person 1</th>
    <th>Person 2</th>
    <th>Person 3</th>
  </tr>
  <tr>
    <td>Emil</td>
    <td>Tobias</td>
    <td>Linus</td>
  </tr>
  <tr>
    <td>16</td>
    <td>14</td>
    <td>10</td>
  </tr>
</table>
```

HTML Table Borders

HTML tables can have borders of different styles and shapes.

How To Add a Border

To add a border, use the CSS *border* property on *table*, *th*, and *td* elements:

Example:

```
table, th, td {
  border: 1px solid black;
}
```

Or use inline css - `<table style="border:1px solid blue">`

Welcome to the VUA-SEN182 CLASS!

Person 1	Person 2	Person 3
Emil	Tobias	Linus
16	14	10

Figure 5: Table with Borders

Collapsed Table Borders

To avoid having double borders, use the `border-collapse` property.

This makes borders appear as a single line.

Example:

```
table, th, td {  
  border: 1px solid black;  
  border-collapse: collapse;  
}
```

Collapsed Borders

Person 1	Person 2	Person 3
Emil	Tobias	Linus

Dotted Table Borders

You can change the border style using the `border-style` property.

Allowed Values:

- dotted
- dashed
- solid
- double
- groove
- ridge
- inset
- outset
- none
- hidden

Example:

```
th, td {  
  border-style: dotted;  
}
```

Welcome to the VUA-SEN182 CLASS!

Person 1	Person 2	Person 3
Emil	Tobias	Linus
16	14	10

Figure 6: Dotted Borders

Border Color

Use the `border-color` property to change border color.

Example:

```
th, td {  
  border-color: #96D4D4;  
}
```

Border Color

Person 1	Person 2	Person 3
Emil	Tobias	Linus

Style Table Borders

You can create an “invisible border” effect using background color.

Example:

```
table, th, td {  
  border: 1px solid white;  
  border-collapse: collapse;  
}
```

```
th, td {  
  background-color: #96D4D4;  
}
```

Invisible Border Style

Person 1	Person 2	Person 3
Emil	Tobias	Linus

Round Table Borders

Use the `border-radius` property to round corners.

Example:

```
table, th, td {
```

```
border: 1px solid black;
border-radius: 10px;
}
```

Rounded Borders

Person 1	Person 2	Person 3
Emil	Tobias	Linus

Without Table Border:

```
th, td {
  border: 1px solid black;
  border-radius: 10px;
}
```

HTML Table Sizes

Tables can have different sizes for rows, columns, or the whole table.

Use the `style` attribute with `width` or `height`.

HTML Table Width

Set the width of a table:

Example:

```
<table style="width:100%">
  <tr>
    <th>Firstname</th>
    <th>Lastname</th>
    <th>Age</th>
  </tr>
  <tr>
    <td>Jill</td>
    <td>Smith</td>
    <td>50</td>
```

```
</tr>
</table>
```

HTML Table Column Width

Table Width 100%

Firstname	Lastname	Age
Jill	Smith	50

Column Width

Set width of a specific column:

Example:

```
<th style="width:70%">Firstname</th>
```

Column Width

Firstname	Lastname	Age
Jill	Smith	50

HTML Table Row Height

Set height of a row:

Example:

```
<tr style="height:200px">
```

Row Height

Firstname	Lastname	Age
Jill	Smith	50

HTML Table Headers

Headers are defined using `<th>` elements.

Example:

```
<table>
  <tr>
    <th>Firstname</th>
    <th>Lastname</th>
    <th>Age</th>
  </tr>
</table>
```

Vertical Table Headers

Use `<th>` in the first column:

Example:

```
<tr>
  <th>Firstname</th>
```

```
<td>Jill</td>
<td>Eve</td>
</tr>
```

Align Table Headers

Headers are bold and centered by default.

To align left:

Example:

```
th {
  text-align: left;
}
```

Header for Multiple Columns

Use `colspan` to span columns.

Example:

```
<th colspan="2">Name</th>
```

HTML Table Padding & Spacing

Cell Padding

Padding is space inside a cell.

Example:

```
th, td {
  padding: 15px;
}
```

Padding

Person 1	Person 2	Person 3
Emil	Tobias	Linus

Individual Padding:

```
th, td {  
  padding-top: 10px;  
  padding-bottom: 20px;  
  padding-left: 30px;  
  padding-right: 40px;  
}
```

Cell Spacing

Spacing is space between cells.

Example:

```
table {  
  border-spacing: 30px;  
}
```


Cell Spacing

Person 1	Person 2	Person 3
Emil	Tobias	Linus

HTML Table Colspan & Rowspan

Colspan

Makes a cell span multiple columns.

Example:

```
<th colspan="2">Name</th>
```

Colspan

Name		Age
Jill	Smith	50

Rowspan

Makes a cell span multiple rows.

Example:

```
<th rowspan="2">Phone</th>
```

Rowspan

Name	Jill
Phone	555-1234
	555-8745

HTML Table Styling

Use CSS to make your tables look better.

HTML Table - Zebra Stripes

If you add a background color on every other table row, you will get a nice zebra stripes effect.

To style every other table row element, use the `:nth-child(even)` selector like this:

Example

```
tr:nth-child(even) {  
  
    background-color: #D6EEEE;  
  
}
```

Zebra Stripes (Horizontal)

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Note: If you use (odd) instead of (even), the styling will occur on row 1,3,5 etc. instead of 2,4,6 etc.

HTML Table - Vertical Zebra Stripes

To make vertical zebra stripes, style every other *column*, instead of every other *row*. Set the `:nth-child(even)` for table data elements like this:

Vertical Zebra Stripes

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Example

```
td:nth-child(even), th:nth-child(even) {  
  
    background-color: #D6EEEE;  
  
}
```

Note: Put the `:nth-child()` selector on both `th` and `td` elements if you want to have the styling on both headers and regular table cells.

Combine Vertical and Horizontal Zebra Stripes

You can combine the styling from the two examples above and you will have stripes on every other row and every other column.

If you use a transparent color you will get an overlapping effect.

Combined Vertical and Horizontal Zebra Stripes

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	16

Use an `rgba()` color to specify the transparency of the color:

Example

```
tr:nth-child(even) {  
  background-color: rgba(150, 212, 212, 0.4);  
}  
th:nth-child(even),td:nth-child(even) {  
  background-color: rgba(150, 212, 212, 0.4);  
}
```

Horizontal Divider

If you specify borders only at the bottom of each table row, you will have a table with horizontal dividers.

Add the `border-bottom` property to all `tr` elements to get horizontal dividers:

Example

```
tr {  
  
  border-bottom: 1px solid #ddd;  
  
}
```

Horizontal Dividers

First Name	Last Name	Savings
Peter	Griffin	\$100
Lois	Griffin	\$150
Joe	Swanson	\$300

Hoverable Table

Use the `:hover` selector on `tr` to highlight table rows on mouse over:

Example

```
tr:hover {background-color: #D6EEEE;}
```

Lab 3: Tables

SECTION B: Lab Tasks

Task 1: Create a Basic HTML Table

What to do

Create a simple table with 3 columns and 3 rows.

Requirements

- Include headings for each column
- Add sample data (names, ages, emails)

Sample Output

Name	Age	Email
Alex Rivera	28	alex.r@example.com
Jordan Smith	34	j.smith@example.com
Sam Tavlör	22	stavlör@example.com

Task 2: Add Table Borders

What to do

Style your table to include visible borders.

Requirements

- Apply borders to table, rows, and cells
- Use inline CSS

Sample Output

Name	Age	Email
Alex Rivera	28	alex.r@example.com
Jordan Smith	34	j.smith@example.com
Sam Taylor	22	staylor@example.com

Task 3: Apply Collapsed Borders

What to do

Modify your table so borders appear as a single line.

Requirement

- Use `border-collapse`

Sample Output

Name	Age	Email
Alex Rivera	28	alex.r@example.com
Jordan Smith	34	j.smith@example.com
Sam Taylor	22	staylor@example.com

Task 4: Add Table Size and Dimensions

What to do

Adjust the width of your table and one column.

Requirements

- Set table width to 100%
- Set first column width to 50%

Sample Output

Student Records

Name	Age	Email
Alex Rivera	28	alex.r@example.com
Jordan Smith	34	j.smith@example.com
Sam Taylor	22	staylor@example.com

Task 5: Style Table with Background Colors

What to do

Add background color to table cells.

Requirement

- Apply color to `<th>` and `<td>`

Suggestion

```
th, td {  
  background-color: #96D4D4;  
}
```

Sample output

Student Records

Name	Age	Email
Alex Rivera	28	alex.r@example.com
Jordan Smith	34	j.smith@example.com
Sam Taylor	22	staylor@example.com

Task 6: Create Zebra Stripes (Horizontal)

What to do

Apply alternating colors to table rows.

Requirement

- Use `nth-child(even)`

Suggestion

```
tr:nth-child(even) {  
  background-color: #D6EEEE;  
}
```

Sample Output

Student Records

Name	Age	Email
Alex Rivera	28	alex.r@example.com
Jordan Smith	34	j.smith@example.com
Sam Taylor	22	staylor@example.com

Task 7: Create Vertical Zebra Stripes

What to do

Apply alternating colors to columns.

Requirement

- Style both `<th>` and `<td>`

Suggestion

```
td:nth-child(even), th:nth-child(even) {  
  background-color: #D6EEEE;  
}
```

Student Records

Name	Age	Email
Alex Rivera	28	alex.r@example.com
Jordan Smith	34	j.smith@example.com
Sam Taylor	22	staylor@example.com

Task 8: Add Cell Padding and Spacing

What to do

Improve spacing inside and between cells.

Requirements

- Add padding
- Add spacing between cells

Suggestion

```
th, td {  
  padding: 10px;  
}
```

```
table {  
  border-spacing: 10px;  
}
```

Sample Output

Student Records

Name	Age	Email
Alex Rivera	28	alex.r@example.com
Jordan Smith	34	j.smith@example.com
Sam Taylor	22	staylor@example.com

Task 9: Create Table with Header Spanning Columns

What to do

Create a table header that spans multiple columns.

Requirement

- Use `colspan`

Suggestion

```
<th colspan="2">Full Name</th>
```

Task 10: Create Table with Row Spanning

What to do

Make a cell span multiple rows.

Requirement

- Use `rowspan`

Suggestion

```
<th rowspan="2">Phone</th>
```

Sample Output for Task 9 and 10

Student Records

Full Name		Name	Age	Email
Phone	Alex Rivera	28	alex.r@example.com	
	Jordan Smith	34	j.smith@example.com	
Sam Taylor	22	staylor@example.com		

Task 11: Add Horizontal Dividers

What to do

Create lines between rows only.

Requirement

- Use `border-bottom`

Suggestion

```
tr {  
  border-bottom: 1px solid #ddd;  
}
```

Student Records

Full Name		Name	Age	Email
Phone	Alex Rivera	28	alex.r@example.com	
	Jordan Smith	34	j.smith@example.com	
Sam Taylor	22	staylor@example.com		

Task 12: Create a Hoverable Table

What to do

Highlight rows when hovered.

Requirement

- Use `:hover`

Suggestion

```
tr:hover {  
  background-color: #D6EEEE;  
}
```

Task 13: Add Rounded Table Borders

What to do

Make table borders rounded.

Requirement

- Use `border-radius`

Suggestion

```
table, th, td {  
  border-radius: 10px;  
}
```

Student Records

Full Name		Name	Age	Email
Phone	Alex Rivera	28	alex.r@example.com	
	Jordan Smith	34	j.smith@example.com	
Sam Taylor	22	staylor@example.com		

Task 14: Create a Complete Styled Table

What to do

Combine multiple styles into one table.

Requirements

- Borders
- Zebra stripes
- Padding
- Hover effect

Instruction

Create a well-formatted student record table demonstrating all the above features.

Sample Output

University Student Records

Student Name	Major	Student Email
Alex Rivera	Computer Science	alex.r@veritas.edu
Jordan Smith	Mechanical Engineering	j.smith@veritas.edu
Sam Taylor	Digital Arts	staylor@veritas.edu
Maria Garcia	Physics	m.garcia@veritas.edu
Liam Chen	Business Admin	l.chen@veritas.edu